

AERO50006 Mathematics 2, Tutorial Sheet 2 (Signals&Systems part)

1. Determine the Fourier series coefficients for each of the following continuous-time periodic signals:

a) $f(t) = e^t, \quad |t| < 1$

b) $f(t) = \begin{cases} \cos(\pi t) & 0 \leq t \leq 3 \\ 0 & 3 \leq t \leq 6 \end{cases}$

$$(a) C_k = \frac{1}{T} \int_T f(t) e^{-jk\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

$$= \frac{1}{2} \int_{-1}^1 e^t e^{-jk\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{2} = \pi$$

$$= \frac{1}{2} \int_{-1}^1 e^{(1-jk\pi)t} dt$$

$$= \frac{1}{2(1-jk\pi)} [e^{(1-jk\pi)t}]_{-1}^1$$

$$= \frac{1}{2(1-jk\pi)} (e^{(1-jk\pi)} - e^{-(1-jk\pi)})$$

$$= \frac{1}{2(1-jk\pi)} (e^1 e^{-jk\pi} - e^{-1} e^{jk\pi})$$

$$= \frac{1}{2(1-jk\pi)} [e^1 (\cos(\pi) - j\sin(\pi))^k - e^{-1} (\cos(\pi) + j\sin(\pi))^k]$$

$$= \frac{1}{2(1-jk\pi)} (e^1 - e^{-1}) (-1)^k$$

$$= \begin{cases} \frac{e^1 - e^{-1}}{2(1-jk\pi)} = \frac{\sinh(1)}{1-jk\pi} & \text{if } k \text{ even} \\ \frac{e^1 - e^{-1}}{2(1-jk\pi)} = \frac{\sinh(1)}{jk\pi - 1} & \text{if } k \text{ odd} \end{cases}$$

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b) $c_k = \frac{1}{6} \int_0^6 f(t) e^{-jk\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{6} = \frac{1}{3}\pi$

$= \frac{1}{6} \int_0^3 \cos(\pi t) e^{-j(\frac{1}{3}k\pi t)} dt$

$\therefore \int_0^3 \cos(\pi t) e^{-j(\frac{1}{3}k\pi t)} dt$

	$\xrightarrow{\text{blue arrow}} \frac{1}{2}(e^{j\pi t} + e^{-j\pi t})$	
	$+ \quad \overset{D}{e^{-j(\frac{1}{3}k\pi t)}} \quad \overset{I}{\cos(\pi t)}$	
$= \frac{1}{\pi} e^{-j(\frac{1}{3}k\pi t)} \sin(\pi t)$	$- \frac{1}{3jk\pi} e^{-j(\frac{1}{3}k\pi t)}$	$\frac{1}{\pi} \sin(\pi t)$
$- \frac{1}{3\pi} jk e^{-j(\frac{1}{3}k\pi t)} \cos(\pi t)$	$+ \frac{1}{3jk\pi} e^{-j(\frac{1}{3}k\pi t)}$	$- \frac{1}{\pi} \cos(\pi t)$
$+ \frac{1}{3jk\pi} \int_0^3 \cos(\pi t) e^{-j(\frac{1}{3}k\pi t)} dt$		

$= \left[\frac{9 \sin(\pi t)}{(9-j^2k^2)\pi} e^{-j(\frac{1}{3}k\pi t)} - \frac{3jk \cos(\pi t)}{(9-j^2k^2)\pi} e^{-j(\frac{1}{3}k\pi t)} \right]_0^3$

$= \frac{3jk}{(9-j^2k^2)\pi} e^{-j(k\pi)} + \frac{3jk}{(9-j^2k^2)\pi} = \frac{3jk}{(9-j^2k^2)\pi} [(-1)^k + 1]$

$\therefore c_k = \begin{cases} j \frac{k}{(9-j^2k^2)\pi} & \text{if } k \text{ even} \\ 0 & \text{if } k \text{ odd} \end{cases}$

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$$\therefore \frac{1}{2} \int_0^3 e^{j(\pi - \frac{1}{3}k\pi)t} + e^{-j(\pi + \frac{1}{3}k\pi)t} dt$$

$$= \frac{j}{2\pi} \left[\frac{3}{3-k} (e^{j\pi(3-k)} - 1) + \frac{3}{3+k} (e^{-j\pi(3+k)} - 1) \right]$$

$$= \dots$$

$$\therefore \begin{cases} C_{-3} = \frac{1}{4} & (k = -3) \\ C_3 = \frac{1}{4} & (k = 3) \\ \dots & (k = \text{general}) \end{cases}$$

2. For each set of Fourier series coefficients below, determine the corresponding continuous-time periodic signal:

a)

$$c_k = \frac{(-1)^k \sin\left(\frac{k\pi}{4}\right)}{k\pi}, \quad c_0 = 0$$

b) $c_k = \begin{cases} k & |k| \leq 2 \\ 0 & \text{otherwise} \end{cases}$

Hint for a): You can make use of the fact that the periodic signal,

$$z(t) = \begin{cases} 1 & \text{if } |t| \leq T_1 \\ 0 & \text{if } T_1 < |t| \leq T/2 \end{cases}$$

has Fourier coefficients $d_0 = \frac{2T_1}{T}$, $d_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$ ($k \neq 0$).

(a) According to the fact list above:

$$x(t) = \begin{cases} (-1)^k & \text{if } |t| \leq \frac{T}{8} \\ 0 & \text{if } \frac{T}{8} < |t| \leq \frac{T}{2} \end{cases}$$

$$\therefore c_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (-1)^k e^{jk\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

$$= \frac{(-1)^k}{T} \int_{-\frac{T}{8}}^{\frac{T}{8}} e^{jk\frac{2\pi}{T}t} dt$$

$$= \frac{(-1)^k}{T} \cdot \frac{T}{2jk\pi} [e^{-jk\frac{2\pi}{T}t}]_{-\frac{T}{8}}^{\frac{T}{8}}$$

$$= \frac{(-1)^k}{k\pi} \cdot \frac{1}{2j} \cdot 2j \sin\left(k\frac{2\pi}{T} \cdot \frac{T}{8}\right) = \frac{(-1)^k \sin\left(\frac{k\pi}{4}\right)}{k\pi}$$

$$c_0 = 0$$

2. For each set of Fourier series coefficients below, determine the corresponding continuous-time periodic signal:

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Hint for a): You can make use of the fact that the periodic signal,

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has Fourier coefficients $d_0 = \frac{2T_1}{T}$, $d_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$ ($k \neq 0$).

$$\text{b) } x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_0 t}$$

$$= \sum_{k=-2}^2 k e^{jk\omega_0 t}$$

$$= -2e^{-2j\omega_0 t} - e^{-j\omega_0 t} + e^{j\omega_0 t} + 2e^{2j\omega_0 t}$$

$$= 2[-\cos(2\omega_0 t) + j\sin(2\omega_0 t) + \cos(2\omega_0 t) + j\sin(2\omega_0 t)]$$

$$+ [-\cos(\omega_0 t) + j\sin(\omega_0 t) + \cos(\omega_0 t) + j\sin(\omega_0 t)]$$

$$= j[4\sin(2\omega_0 t) + 2\sin(\omega_0 t)]$$

$$= j[8\sin(\omega_0 t)\cos(\omega_0 t) + 2\sin(\omega_0 t)]$$

$$\text{where } \omega_0 = \frac{2\pi}{T}$$

3. Show that the Fourier transform of the signal

$$x(t) = \begin{cases} e^{-at} & t \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0$, is

$$X(j\omega) = \frac{1}{(a + j\omega)}.$$

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt \quad \text{where } \omega = \frac{2\pi k}{T}$$

$$= \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \frac{-1}{a+j\omega} [e^{-(a+j\omega)t}]_0^{\infty} \quad (a > 0)$$

$$= \frac{-1}{a+j\omega} (0 - 1)$$

$$= \frac{1}{a+j\omega}$$

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4. Show that if the continuous-time signal $x(t)$ has Fourier transform $X(j\omega)$, then

$$-jtx(t) \xleftrightarrow{\mathcal{F}} \frac{dX(j\omega)}{d\omega}.$$

Hence show that the Fourier transform of

$$x(t) = \begin{cases} \frac{t^{n-1}}{(n-1)!} e^{-at} & t \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0$, is

$$X(j\omega) = \frac{1}{(a + j\omega)^n}.$$

$$\text{Let } \hat{F}(j\omega) = \frac{d\hat{X}(j\omega)}{d\omega}$$

$$\begin{aligned} \therefore f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d\hat{X}(j\omega)}{d\omega} e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \left(\left[\hat{X}(j\omega) e^{j\omega t} \right]_{-\infty}^{\infty} \right. \\ &\quad \left. - \frac{j t}{2\pi} \int_{-\infty}^{\infty} \hat{X}(j\omega) e^{j\omega t} d\omega \right) \\ &= -j t x(t) \end{aligned}$$

$$\therefore \text{for } y(t) = \begin{cases} e^{-at} & t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\hat{Y}(j\omega) = \frac{1}{a+j\omega}$$

$$\therefore x(t) = \frac{t^{n-1}}{(n-1)!} y(t)$$

$$\begin{aligned} \therefore \hat{X}(j\omega) &= \int_{-\infty}^{\infty} \frac{t^{n-1}}{(n-1)!} y(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} \frac{t^{n-1}}{(n-1)!} e^{-(a+j\omega)t} dt \quad + \quad \frac{t^{n-1}}{(n-1)!} e^{-(a+j\omega)t} \\ &= \frac{1}{(n-1)!} \int_0^{\infty} t^{n-1} e^{-(a+j\omega)t} dt \quad - \quad (n-1) t^{n-2} \frac{-1}{a+j\omega} e^{-(a+j\omega)t} \\ &= \frac{1}{(n-1)!} \int_0^{\infty} t^{n-1} e^{-(a+j\omega)t} dt \quad \dots \quad \frac{(-1)^{n-1}}{(a+j\omega)^{n-1}} e^{-(a+j\omega)t} \\ &= \dots + \frac{1}{(n-1)!} \left[(-1)^n (n-1)! + \frac{(-1)^{n-1}}{(a+j\omega)^{n-1}} e^{-(a+j\omega)t} \right]_0^{\infty} \\ &= \frac{1}{(n-1)!} \int_0^{\infty} (-1)^{n-1} (n-1)! \frac{(-1)^{n-1}}{(a+j\omega)^{n-1}} e^{-(a+j\omega)t} dt \\ &= \frac{1}{(a+j\omega)^n} \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\ &= \frac{1}{(a+j\omega)^n} \hat{Y}(j\omega) \\ &= \frac{1}{(a+j\omega)^n} \end{aligned}$$

5. Suppose that the input $x(t)$ and output $y(t)$ of the LTI system H are related by:

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = 2x(t).$$

- Find the frequency response, $H(j\omega)$, of this system.
- Find the impulse response of the system.
- Find the response of the system to the input

$$x(t) = \begin{cases} te^{-at} & t \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0$.

- Find the response of the system to the input

$$x(t) = 2 \cos(3t).$$

$$a) [(j\omega)^2 + 6j\omega + 8] Y(j\omega) = 2X(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2}{(j\omega)^2 + 6j\omega + 8} = \frac{2}{(j\omega + 2)(j\omega + 4)}$$

$$= \frac{1}{j\omega + 2} - \frac{1}{j\omega + 4}$$

$$b) h(t) = (e^{-2t} - e^{-4t}) u(t)$$

$$c) y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_0^{\infty} \tau e^{-a\tau} (e^{-2(t-\tau)} - e^{-4(t-\tau)}) d\tau$$

$$= \int_0^{\infty} \tau e^{-2t} e^{(2-a)\tau} - \tau e^{-4t} e^{(4-a)\tau} d\tau$$

$$\begin{array}{rcl} 0 & & 1 \\ + & \tau & e^{(2-a)\tau} \\ - & 1 & \frac{1}{2-a} e^{(2-a)\tau} \\ + & 0 & \frac{1}{(2-a)^2} e^{(2-a)\tau} \end{array}$$

$$= e^{-2t} \left[\left(\frac{\tau}{2-a} - \frac{1}{(2-a)^2} \right) e^{(2-a)\tau} \right]_0^{\infty} - e^{-4t} \left[\left(\frac{\tau}{4-a} - \frac{1}{(4-a)^2} \right) e^{(4-a)\tau} \right]_0^{\infty}$$

$$= \frac{e^{-2t}}{(2-a)^2} - \frac{e^{-4t}}{(4-a)^2}$$

$$\text{OR } \hat{Y}(j\omega) = \hat{X}(j\omega) \cdot \hat{H}(j\omega)$$

$$(d) y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= 2 \int_{-\infty}^{\infty} \cos(3\tau) (e^{-2(t-\tau)} - e^{-4(t-\tau)}) d\tau$$

$$= 4 \int_0^{\infty} \cos(3t-3\tau) (e^{-2\tau} - e^{-4\tau}) d\tau$$

$$= 4 \int_0^{\infty} \cos[3(t-\tau)] e^{-2\tau} d\tau - 4 \int_0^{\infty} \cos[3(t-\tau)] e^{-4\tau} d\tau$$

$\begin{array}{rcl} & 0 & 1 \\ + & \cos[3(t-\tau)] & e^{-2\tau} \\ - & 3\sin[3(t-\tau)] & -\frac{1}{2}e^{-2\tau} \\ + & -9\cos[3(t-\tau)] & \frac{1}{4}e^{-2\tau} \end{array}$	$\begin{array}{rcl} & 0 & 1 \\ + & \cos[3(t-\tau)] & e^{-4\tau} \\ - & 3\sin[3(t-\tau)] & -\frac{1}{4}e^{-4\tau} \\ + & -9\cos[3(t-\tau)] & \frac{1}{16}e^{-4\tau} \end{array}$
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$$= 4 \left[-\frac{2\cos[3(t-\tau)] + 3\sin[3(t-\tau)]}{13} e^{-2\tau} \right]_0^{\infty}$$

$$- 4 \left[-\frac{4\cos[3(t-\tau)] + 3\sin[3(t-\tau)]}{25} e^{-2\tau} \right]_0^{\infty}$$

$$= 4 \left[\frac{2}{13} \cos(3t) + \frac{3}{13} \sin(3t) - \frac{4}{25} \cos(3t) - \frac{3}{25} \sin(3t) \right]$$

$$= \left(\frac{8}{13} - \frac{16}{25} \right) \cos(3t) + \left(\frac{12}{13} - \frac{12}{25} \right) \sin(3t)$$